PHYSICAL EXPLANATIONS FOR PARAMETER ASSIGNMENTS

A structured reference for tuning a wave-field / CSNF-style model across dynamical regimes

Purpose
This note organizes the proposed parameter-selection heuristics and numerical safeguards into a concise, implementation-oriented reference. Parameter symbols are made explicit where they were absent in the supplied text; map them to the names used in your implementation.

1. Physical explanations by dynamical regime

1.1 Continuous chaotic attractors

Examples: Lorenz system and driven pendulum

Parameter rule: high spatial resolution N, narrow packet width σ, and low readout regularization λ.

• Resolve attractor lobes. Chaotic attractors can place nearby states on trajectories with sharply different futures. Broad Gaussian packets overlap across distinct regions of the ring and blur their representations; narrower packets can preserve more local distinctions.

• Exploit the feature geometry carefully. Rich trajectories may produce diverse feature vectors, but linear independence is not guaranteed. Keep λ low only when validation indicates that the readout is well-conditioned; otherwise regularization remains useful.

1.2 Discrete chaotic maps

Example: logistic map

Parameter rule: high input-injection strength η and moderate diffusion D.

• Reduce unwanted temporal inertia. A discrete map advances by iteration rather than by an explicitly modelled continuous momentum state. Excessive recurrent retention can introduce hysteresis, so stronger injection lets the current input exert more influence.

• Use dispersion to shape features. Moderate diffusion, together with spatial potential structure, can broaden and deform packets into smoother nonlinear basis patterns. This is a modelling hypothesis to test empirically, not a guarantee that the field exactly represents a quadratic map.

1.3 Delay differential equations

Example: Mackey–Glass

Parameter rule: high memory retention ρ, high grid resolution N, and active carrier dispersion / circulation ω.

• Represent delayed dependence. Delay differential equations depend on a history segment, not only the present state. A persistent field can act as a distributed memory, while a finer ring can represent more history detail.

• Couple stored and incoming information. Carrier dynamics may move or mix information across spatial locations, potentially coupling older state with new input. The analogy to a ring buffer is conceptual: a recurrent field is not automatically an exact delay line, so test delay reconstruction and stability directly.

1.4 Smooth conservative limit cycles

Example: Lotka–Volterra

Parameter rule: tune carrier frequency ω to the orbit period / dominant frequency, use a relatively low grid dimension N, broad packets σ, and moderate diffusion D.

- Match temporal scales. If the carrier oscillates much faster than the observed orbit, the readout may need to cancel unnecessary high-frequency structure, potentially harming phase accuracy. Frequency matching is a sensible starting point; validate it against held-out trajectories.
- Prefer smooth distributed modes. Moderate diffusion can spread localized packets into smoother, harmonic-like patterns. With no diffusion, the representation may remain overly localized; excessive diffusion can instead erase useful structure.

1.5 Stochastic, spiky, and burst series

Example: sunspot time series

Parameter rule: very low diffusion D, high spatial dimensionality N, high carrier frequency ω, and low memory retention ρ.

- Preserve abrupt local events. Low diffusion limits spatial spreading, which may help retain localized bursts rather than smear them across neighboring bins. The “solitonic particle” description is an analogy unless the implemented dynamics establish soliton behaviour.
- Limit persistence of stale noise. In a recurrent field, high retention can carry old noise forward. Lower ρ can flush stale information more quickly, but may also discard useful temporal structure. Tune against time-ordered validation data and compare with simple baselines.

2. Three invariant numerical rules

These are implementation safeguards. They should be applied consistently unless the mathematical formulation is deliberately changed and re-validated.

Rule 1 — Analytical centering instead of bias augmentation

Do not append a constant 1.0 feature merely to represent the intercept when using the stated centered-feature formulation. Center the feature matrix and target using training-set statistics, fit the readout, and compute the intercept analytically.

$$\tilde{X} = X - \mu_x \; ; \; \tilde{y} = y - \mu_Y$$
$$b = \mu_Y - \mu_x^T w$$

Technical note: Important qualification: probability conservation by a unitary propagator does not, by itself, prove that adding a bias feature creates an exact zero eigenvalue. The prohibition is a design rule for this formulation, not a universal theorem about unitary systems.

Rule 2 — Geodesic / shortest-arc velocity for angles

For cyclic variables, never calculate angular velocity using unwrapped raw subtraction. Use the signed shortest-arc difference so a transition across the periodic boundary does not create a spurious jump.

$$\Delta\theta = \text{atan2}(\sin(\theta_t - \theta_{t-1}), \cos(\theta_t - \theta_{t-1}))$$
$$v\theta = \Delta\theta / \Delta t$$

Technical note: This returns a wrapped angular difference in the interval $(-\pi, \pi]$, assuming radians. Confirm units and handle the exact antipodal case consistently.

Rule 3 — Truncated SVD for smooth / ill-conditioned readouts

Solve the linear readout with a truncated singular-value decomposition rather than directly inverting the feature covariance or Gram matrix. Use the stated relative cutoff rcond = 1e−4 as the initial policy.

$$\tilde{X} = U\Sigma V^T \; ; \; \sigma_i \text{ is retained when } \sigma_i / \sigma\text{max} > 10^{-4}$$

Technical note: Smooth signals can make neighboring spatial-bin features highly correlated. Truncation suppresses poorly identified directions and reduces coefficient blow-up; it may introduce bias, so report the cutoff and evaluate sensitivity.

3. Quick-reference parameter matrix

Dynamical regime	Suggested starting point	Primary rationale
Continuous chaos	N ↑ · σ ↓ · λ low	Resolve local attractor structure; avoid unnecessary smoothing
Discrete chaotic maps	η ↑ · D moderate	Let current input dominate; shape nonlinear spatial features
Delay differential equations	ρ ↑ · N ↑ · ω active	Retain and circulate history information
Smooth limit cycles	ω matched · N modest · σ broad · D moderate	Match the orbit timescale and favour smooth modes

Dynamical regime	Suggested starting point	Primary rationale
Spiky / burst series	D very low · N high · ω high · ρ low	Localize bursts and limit persistence of stale noise

4. Recommended validation protocol

- Treat the parameter rules as initial hypotheses, not universal optima. Tune on a time-ordered validation split and preserve a final untouched test set.
- Report sensitivity to grid size, packet width, diffusion, retention, carrier frequency, regularization, and SVD cutoff.
- For chaotic and stochastic systems, evaluate across multiple seeds or temporal windows; report both average performance and variability.
- Inspect numerical conditioning, readout weight norms, phase error for cyclic systems, and forecast stability—not only mean-squared error.

Notation key: N = spatial/grid resolution; σ = packet width; λ = readout regularization; η = input injection; D = diffusion; ρ = memory retention; ω = carrier frequency or circulation parameter; μ_x and μ_y = training-set feature and target means.

Scientific-use note: the physical explanations above are mechanistic interpretations of the proposed architecture. They should be presented as hypotheses unless supported by derivations or controlled ablation experiments.